$$\frac{dy}{dx} = \sqrt{2x+3}$$

$$\Rightarrow \quad dy = \sqrt{2x+3}\ dx.$$

Integrating both sides.

$$\int dy = \int \sqrt{2x+3}\ dx.$$

$$\Rightarrow \quad y = \int \sqrt{u} \cdot \frac{du}{2}$$

$$= \frac{1}{2} \int u^{1/2}\ du.$$

$$= \frac{1}{2} \cdot \frac{u^{1/2+1}}{\frac{1}{2}+1} + c$$

$$= \frac{1}{2} \cdot \frac{u^{3/2}}{3/2} + c$$

$$= \frac{1}{2} \times \frac{2}{3} \cdot u^{3/2} + c$$

$$y(x) = \frac{1}{3} \cdot (2x+3)^{3/2} + c$$

$u = 2x+3.$

$du = 2 \cdot dx.$

$dx = \frac{du}{2}$

① $\int K \cdot f(x)\,dx = K \int f(x)\,dx$, $K \neq 0$.

② $\int \left(f(x) + g(x)\right) dx = \int f(x)\,dx + \int g(x)\,dx$.

③ $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + c \quad (n \neq -1)$

④ $\int \dfrac{1}{x}\,dx = \ln|x| + c$.

⑤ $\int \cos x\,dx = \sin x + c$.

⑥ $\int \sin x\,dx = -\cos x + c$.

⑦ $\int \tan x\,dx = \ln|\sec x| + c$.

⑧ $\int \cot x\,dx = \ln|\sin x| + c$.

$\tan x = \dfrac{\sin x}{\cos x}$

$\cot x = \dfrac{\cos x}{\sin x}$.

9. $\int \sec^2 x \, dx = \tan x + c.$

10. $\int \csc^2 x \, dx = -\cot x + c.$

11. $\int \sec x \tan x \, dx = \sec x + c.$

12. $\int \csc x \cot x \, dx = -\csc x + c.$

13. $\int e^x \, dx = e^x + c.$

14. $\int a^x \, dx = \dfrac{a^x}{\ln a} + c. \quad (a \neq 1, a > 0)$

15. $\int \dfrac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + c$

$\sin x = a$

$x = \sin^{-1} a.$

16. $\int \dfrac{dx}{1+x^2} = \tan^{-1} x + c$

17. $\int \dfrac{dx}{x \cdot \sqrt{x^2-1}} = \sec^{-1} |x| + c.$

① $$\int 5x\,dx = \frac{5x^2}{2} + c.$$

② $$\int \left(x^4 + \sqrt[3]{x^2} - \frac{2}{x^2} - \frac{1}{3\sqrt[3]{x}}\right) dx.$$

$$= \int x^4\,dx + \int \sqrt[3]{x^2}\,dx - \int \frac{2}{x^2}\,dx - \int \frac{1}{3\sqrt[3]{x}}\,dx$$

$$x^{2/3}$$

$$= \frac{x^5}{5} + \frac{x^{5/3}}{5/3} - \frac{2x^{-1}}{-1} - \frac{1}{3}\cdot\frac{x^{2/3}}{2/3} + c$$

$$= \frac{x^5}{5} + \frac{3}{5}x^{5/3} + \frac{2}{x} - \frac{1}{2}x^{2/3} + c.$$

---

③ $$\int 3 - 4x + 2x^3\,dx$$

$$= 3\int 1\cdot dx - 4\cdot\int x\,dx + 2\int x^3\,dx$$

$$= 3x - 4\cdot\frac{x^2}{2} + 2\cdot\frac{x^4}{4} + c$$

$$= 3x - 2x^2 + \frac{1}{2}x^4 + c.$$

---

④ $\int 2(1-3x)^2\,dx.$

Method of substitution :

Let $u = 1-3x$

$du = -3\,dx \Rightarrow dx = -\frac{1}{3}\,du.$

$2\cdot\int u^2\left(-\frac{1}{3}\,du\right)$

$= -\frac{2}{3}\int u^2\,du = -\frac{2}{3}\cdot\frac{u^3}{3} + c.$

$= -\frac{2}{9}(1-3x)^3 + c$

---

⑤ $\int (2x^3-1)^5\cdot x^2\,dx.$

$= \int u^5\cdot \cancel{x^2}\cdot\frac{du}{6\cancel{x^2}}$

$= \frac{1}{6}\int u^5\cdot du$

$= \frac{1}{6}\cdot\frac{u^6}{6} + c.$

$\hookrightarrow = \frac{(2x^3-1)^6}{36} + c.$

$u = 2x^3-1$

$\Rightarrow du = 2\cdot(3x^2)\,dx$

$\Rightarrow du = 6x^2\,dx.$

$\frac{d}{dx}(x^n) = n\cdot x^{n-1}$

$\frac{d}{dx}(x^3) = 3x^2$